

1. Introduction to solutions

A solution is a homogeneous mixture of two or more components whose composition can vary within limits.^[1]

The component present in larger amount (or deciding the physical state) is called the solvent; the other components are solutes.^[1]

- **Binary solution:** one solute + one solvent.
- **Multicomponent solution:** more than one solute in a common solvent.

According to phases of solute and solvent, examples include:

- Gas in gas (air), gas in liquid (CO₂ in water), liquid in liquid (ethanol–water), solid in liquid (salt–water), solid in solid (alloys like brass).^[1]

2. Types of solutions by concentration

At a given temperature:

- **Unsaturated solution:** contains less solute than required for equilibrium solubility; more solute can dissolve.
- **Saturated solution:** in equilibrium with undissolved solute; rate of dissolution = rate of crystallisation.^[1]
- **Supersaturated solution:** contains more solute than solubility; metastable and crystallises on slight disturbance.

Concentration can be expressed in many ways; choice depends on context.

3. Concentration terms

3.1 Mass percentage, volume percentage, ppm

1. **Mass % (w/w):**

$$\% \text{ by mass} = \frac{\text{mass of solute}}{\text{mass of solution}} \times 100$$

Used when masses are convenient (e.g., lab preparations).^[1]

2. **Volume % (v/v)** (for liquid–liquid):

$$\% \text{ by volume} = \frac{\text{volume of solute}}{\text{volume of solution}} \times 100^[1]$$

3. **Mass by volume % (w/v)**:

$$\% (w/v) = \frac{\text{mass of solute in g}}{\text{volume of solution in mL}} \times 100$$

Common in medicine (0.9% saline, etc.).^[1]

4. **ppm (parts per million)**:

$$\text{ppm} \approx \frac{\text{mass of solute}}{\text{mass of solution}} \times 10^6$$

Used for very dilute solutions like pollutants in water/air.^[1]

3.2 Mole fraction

For a component i in a mixture:

$$x_i = \frac{n_i}{\sum n_j}$$

where n_i is moles of that component.^[1]

- Dimensionless, temperature independent.
- In a binary solution: $x_1 + x_2 = 1$.
- Very convenient in vapour-pressure and colligative-property formulae.^[1]

3.3 Molarity (M) and molality (m)

1. **Molarity (M)**:

$$M = \frac{\text{moles of solute}}{\text{volume of solution in L}}$$

Depends on temperature (volume changes).^[1]

2. Molality (m):

$$m = \frac{\text{moles of solute}}{\text{mass of solvent in kg}}$$

Independent of temperature; preferred in colligative-property work.^[1]

For dilute aqueous solutions with density $\approx 1 \text{ g mL}^{-1}$, M and m are numerically close, but conceptually different.^[1]

4. Solubility and factors affecting it

4.1 Solubility definition

Solubility of a solute is the maximum amount that can be dissolved in a given amount of solvent at a specified temperature and pressure, giving a saturated solution.^[1]

Units: g solute per 100 g solvent, or mol L^{-1} , etc.

4.2 Solids in liquids

For most ionic and polar solids in water:

- Solubility generally **increases with temperature** if dissolution is endothermic.
- If dissolution is exothermic, solubility can **decrease with temperature**.^[1]

Stirring and smaller particle size speed up dissolution but do not change equilibrium solubility.

4.3 Gases in liquids and Henry's law

Gas solubility in liquids:

- **Increases with pressure.**
- **Decreases with temperature** (for most gases in water).^{[2][1]}

Henry's Law: At constant temperature, solubility of a gas in a liquid is directly proportional to partial pressure of the gas above the solution.^{[2][1]}

One common form:

$$p = k_H x$$

where p is partial pressure of gas in vapour phase, x is mole fraction of gas in solution, k_H is Henry's constant.^[1]

- Larger $k_H \Rightarrow$ lower solubility (for fixed p).^[2]
- k_H increases with T, so solubility usually decreases with T.^[1]

Applications:

- Bottling of soda/soft drinks at high pressure, low temperature.^[2]
- Explaining “bends” in deep-sea divers (N_2 coming out of blood when pressure suddenly decreases).^[2]

Henry's law is approximate; strong solute–solvent interactions and very high pressures cause deviations.^[1]

5. Ideal and non-ideal solutions, Raoult's law

5.1 Raoult's law for solvent

For a solution of a non-volatile solute in a volatile solvent:

- **Raoult's law:** Partial vapour pressure of solvent above solution is proportional to its mole fraction in solution.^{[3][1]} $p_{\text{solvent}} = x_{\text{solvent}} p_{\text{solvent}}^{\circ}$

where p° is vapour pressure of pure solvent.

Relative lowering of vapour pressure is:

$$\frac{p^{\circ} - p}{p^{\circ}} = x_{\text{solute}}$$

for dilute solutions with non-volatile solute.^[1]

5.2 Ideal solutions (volatile–volatile)

For a binary mixture of volatile liquids A and B:

$$p_A = x_A p_A^\circ, p_B = x_B p_B^\circ, p_{\text{total}} = p_A + p_B \quad [1]$$

An **ideal solution** shows:

- Obedience to Raoult's law over entire composition range.
- $\Delta H_{\text{mix}} = 0$ (no heat absorbed or evolved on mixing).
- $\Delta V_{\text{mix}} = 0$ (no volume change on mixing).^[1]

Approximate examples: benzene–toluene, n-hexane–n-heptane (similar size and interactions).^[1]

5.3 Non-ideal solutions and deviations

Real solutions often deviate from Raoult's law:

- **Positive deviation:**
 - Vapour pressure higher than Raoult's prediction.
 - A–B interactions weaker than A–A, B–B; molecules escape more easily.^{[3][1]}
 - $\Delta H_{\text{mix}} > 0, \Delta V_{\text{mix}} > 0$.
 - Example: ethanol–water (H-bond network partially broken → more escaping tendency).^[3]
- **Negative deviation:**
 - Vapour pressure lower than predicted.
 - A–B interactions stronger than A–A, B–B (extra attraction).^{[3][1]}
 - $\Delta H_{\text{mix}} < 0, \Delta V_{\text{mix}} < 0$.
 - Example: acetone–chloroform, HCl–water (strong specific interactions like H-bonding or complex formation).^[3]

5.4 Azeotropes

An **azeotrope** is a constant-boiling mixture whose vapour composition equals liquid composition at a certain composition and pressure.^[1]

- **Minimum-boiling azeotrope:** arises from positive deviation; total vapour pressure maximum, boiling point minimum (ethanol–water).
- **Maximum-boiling azeotrope:** arises from negative deviation; total vapour pressure minimum, boiling point maximum (e.g., nitric acid–water, some HCl–water compositions).^{[3][1]}

Azeotropic mixtures cannot be completely separated into pure components by simple distillation.

6. Colligative properties – overview

Colligative properties are properties of dilute solutions that depend only on the number of solute particles relative to solvent particles, not on their nature.^{[4][2][1]}

The four colligative properties are:

1. Relative lowering of vapour pressure.
2. Elevation of boiling point.
3. Depression of freezing point.
4. Osmotic pressure.^{[4][1]}

These are strictly colligative in the dilute-solution (near-ideal) limit.^[1]

7. Relative lowering of vapour pressure

For a non-volatile solute in volatile solvent:

$$\frac{p^\circ - p}{p^\circ} = x_{\text{solute}} \quad [1][5]$$

In terms of moles:

$$\frac{p^\circ - p}{p^\circ} = \frac{n_{\text{solute}}}{n_{\text{solute}} + n_{\text{solvent}}} \approx \frac{n_{\text{solute}}}{n_{\text{solvent}}} \quad (\text{dilute}) \quad [1]$$

- Adding non-volatile solute **lowers vapour pressure**, because some surface sites are occupied by solute particles.^[6]
- This lowering is the basis for boiling-point elevation and freezing-point depression.

8. Elevation of boiling point

Boiling point is the temperature at which vapour pressure equals external pressure.^[6]

When a non-volatile solute is added:

- Vapour pressure at a given temperature decreases.
- To reach external pressure, solution must be heated more $\Rightarrow$ **boiling point increases.**^{[5][6]}

For a dilute solution:

$$\Delta T_b = T_b - T_b^\circ = K_b m$$

where:

- K_b is molal elevation constant (ebullioscopic constant), depends only on solvent.
- m is molality of solute.^{[4][1]}

Uses:

- Determination of molar mass of non-volatile solutes from measured ΔT_b .^[1]

9. Depression of freezing point

Freezing point is the temperature at which solid and liquid phases of solvent are in equilibrium.^[6]

Adding non-volatile solute:

- Lowers the chemical potential of liquid phase more than that of solid.
- Equilibrium occurs at lower T $\Rightarrow$ **freezing point decreases.**^{[2][1]}

For dilute solutions:

$$\Delta T_f = T_f^\circ - T_f = K_f m$$

where:

- K_f is molal depression constant (cryoscopic constant) of solvent.
- m is molality.^{[4][1]}

Uses:

- Widely used in determining molar mass of solutes; often more sensitive than ΔT_b .^{[4][2]}
- Basis of antifreeze in car radiators and salting of roads (lowering freezing point of water).

10. Osmosis and osmotic pressure

Osmosis: Flow of solvent through a semipermeable membrane from a region of lower solute concentration (more pure solvent) to higher solute concentration.^{[6][1]}

- **Semipermeable membrane:** allows only solvent molecules, not solute.^[1]
- Flow continues until hydrostatic pressure difference builds up and equilibrium is reached.

Osmotic pressure (π): The minimum pressure that must be applied on the solution side to stop osmosis.^{[6][1]}

For dilute ideal solutions:

$$\pi = CRT$$

where C is molarity, R gas constant, T absolute temperature.^{[6][1]}

In moles and volume:

$$\pi = \frac{n}{V}RT$$

This relation is analogous to ideal gas equation; thus osmotic pressure is clearly colligative (depends on concentration of solute particles).^{[6][1]}

Applications:

- Molar mass determination of macromolecules (proteins, polymers) because even very dilute solutions give measurable π .^{[7][1]}
- Biological systems: isotonic, hypertonic, hypotonic solutions and cell behaviour (swelling/shrinking).^[2]
- Reverse osmosis: applying pressure $> \pi$ on solution side to force solvent back through membrane (used in desalination).^{[7][2]}

11. Van't Hoff factor and abnormal molar mass

If solute undergoes association (dimerisation, etc.) or dissociation (electrolytes), effective number of particles changes and colligative properties show **abnormal** values.^{[2][1]}

Van't Hoff factor i :

$$i = \frac{\text{actual number of particles in solution}}{\text{number expected for nonelectrolyte}}$$

or equivalently:

$$i = \frac{\text{observed colligative property}}{\text{calculated colligative property (nonelectrolyte)}} \cdot \square\square$$

- For **dissociation**: $i > 1$ (more particles).
- For **association**: $i < 1$ (fewer particles).^{[4][1]}

Colligative formulas are modified as:

- $\Delta T_b = iK_b m$
- $\Delta T_f = iK_f m$
- $\pi = iCRT$
- $\frac{p^\circ - p}{p^\circ} = i \frac{n_{\text{solute}}}{n_{\text{solvent}}}$ (in appropriate form).^{[2][1]}

From i , one can calculate degree of dissociation or association.

12. Summary of key formulas

For dilute solutions of non-volatile solute:

- **Relative lowering of vapour pressure:**

$$\frac{p^\circ - p}{p^\circ} = x_{\text{solute}} \cdot [1]$$

- **Elevation of boiling point:**

$$\Delta T_b = iK_b m \cdot [4][1]$$

- **Depression of freezing point:**

$$\Delta T_f = iK_f m. [2][11]$$

- **Osmotic pressure:**

$$\pi = iCRT. [6][11]$$

These relations underpin almost all numerical questions on the chapter.

If you want, a separate sheet can be built with fully worked examples (step-by-step) for each formula and typical JEE question patterns (finding molar mass, degree of dissociation, behaviour of azeotropes, etc.).

1. https://en.wikipedia.org/wiki/Colligative_properties
2. <https://byjus.com/jee/colligative-properties/>
3. <https://allen.in/jee/chemistry/colligative-properties>
4. <https://www.vedantu.com/chemistry/colligative-properties>
5. <https://www.chem.fsu.edu/chemlab/chm1046course/colligative.html>
6. <https://chemed.chem.purdue.edu/genchem/topicreview/bp/ch15/colligative.php>
7. <https://ecampusontario.pressbooks.pub/queenschem1/chapter/chapter-11-colligative-properties/>
8. https://www.shaalaa.com/question-bank-solutions/define-term-colligative-properties_16667
9. [https://chem.libretexts.org/Bookshelves/Physical_and_Theoretical_Chemistry_Textbook_Maps/Supplemental_Modules_\(Physical_and_Theoretical_Chemistry\)/Physical_Properties_of_Matter/Solutions_and_Mixtures/Colligative_Properties](https://chem.libretexts.org/Bookshelves/Physical_and_Theoretical_Chemistry_Textbook_Maps/Supplemental_Modules_(Physical_and_Theoretical_Chemistry)/Physical_Properties_of_Matter/Solutions_and_Mixtures/Colligative_Properties)
10. <https://www.youtube.com/watch?v=c8dDLe37ONg>